THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · HARD LEVEL UNIT TEST SET 2

Chapters 5–8: **Parallel and Intersecting Lines, Number Play, A Tale of Three Intersecting Lines, Working with Fractions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. $\frac{1}{3}$ of a wall is already painted. Then $\frac{1}{2}$ of the remaining part is painted. The fraction of the whole wall still unpainted is: 1

(a) $\frac{1}{6}$ **(b)** $\frac{2}{3}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{7}{12}$

2. Two complementary angles differ by 20°. The larger angle is: 1

(a) 65° **(b)** 45° **(c)** 75° **(d)** 55°

3. For every whole number n, $n(n+1)$ is: 1

(a) even **(b)** a perfect square **(c)** odd **(d)** a multiple of 3

4. Sides 11 cm, 6 cm and 5 cm are given. Which statement is true? 1

(a) a triangle can be formed **(b)** only a right triangle is possible

(c) no triangle can be formed **(d)** only an equilateral triangle is possible

5. If $0 < x < 1$, then $\frac{1}{x}$ is: 1

(a) less than 1 **(b)** greater than 1 **(c)** negative **(d)** equal to 1

6. Two angles of a linear pair differ by 20°. The smaller angle is: 1

(a) 70° **(b)** 90° **(c)** 100° **(d)** 80°

7. If $1 + 2 + 3 + \cdots + n = 231$, then n is: 1

(a) 21 **(b)** 23 **(c)** 20 **(d)** 22

8. In an isosceles triangle with perimeter 34 cm the equal sides are 12 cm each. The base (cm) is: 1

(a) 12 **(b)** 11 **(c)** 9 **(d)** 10

9. $\frac{3}{5}$ of $3\frac{1}{3}$ is: 1

(a) 4 **(b)** 2 **(c)** 1 **(d)** $3\frac{14}{15}$

10. One angle of a linear pair is 2 times the other. The larger angle is: 1

(a) 140° **(b)** 110° **(c)** 130° **(d)** 120°

11. The number of reverse-and-add steps for 59 to become a palindrome is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 5

12. The two acute angles of a right triangle are in the ratio $1 : 2$. The smaller acute angle is: 1

(a) 50° **(b)** 20° **(c)** 40° **(d)** 30°

13. **Assertion (A):** $\frac{2}{3} \times \frac{3}{4} = \frac{1}{2}$. 1
Reason (R): The product of two fractions is (product of numerators) / (product of denominators).
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

14. **Assertion (A):** The two angles of a linear pair are supplementary. 1
Reason (R): Vertically opposite angles are equal.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

15. **Assertion (A):** 5832 is divisible by 9. 1
Reason (R): 5832 is an even number.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Show that $n(n+1)$ is even, and verify for $n = 13$. 2

17. In isosceles $\triangle ABC$ with $AB = AC$, side BC is extended to D and $\angle ACD = 115^\circ$. Find $\angle A$ and $\angle B$. 2

18. After $\frac{2}{7}$ of a number is removed, 35 is left. Find the number and the part removed. 2

19. Two supplementary angles differ by 18°. Find both angles and the complement of the smaller one. 2

20. Using each of the digits $7, 3, 4, 1$ once, make the largest 4-digit number divisible by 3. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A cyclist rides at $\frac{9}{2}$ km per hour for 2 hours, then at $\frac{15}{4}$ km per hour for 4 hours. Find the total distance and the average speed for the whole journey. 3

22. Two adjacent angles form a linear pair and one of them is 40°. Find the other angle, the half of each angle and the angle between the two bisectors. 3

23. In how many ways can 15 be written as the sum of two or more consecutive positive whole numbers? List them. 3

24. Two sides of a triangle are 8 cm and 21 cm. The third side is a whole number of cm divisible by 5. List all possible lengths and the largest perimeter. 3

OR

A pole of 20 m is cut into pieces of $1\frac{2}{3}$ m each. How many pieces are obtained and how much is left? 3

25. Angles on one side of a straight line at a point are 20°, 75°, 30° and a fourth angle. Find the fourth angle and name the kind of each angle. 3

OR

The number $7 * 2 * 1$ has the same digit $*$ in both blanks and is divisible by 9. Find all possible values of $*$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** Teacher Neha shows how 15 can be written as a sum of consecutive numbers: $15 = 7 + 8 = 4 + 5 + 6 = 1 + 2 + 3 + 4 + 5$. 5

(i) Which of these sums uses exactly three numbers?

(ii) Write 21 as the sum of three consecutive numbers.

(iii) Is the sum of three consecutive numbers always divisible by 3? Give the reason.

(iv) Write 33 as the sum of three consecutive numbers and also as the sum of two consecutive numbers.

27. $\triangle ABC \cong \triangle DEF$ with $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$. Given $\angle A = (3x + 10)^\circ$, $\angle D = (5x - 20)^\circ$, $AB = (2y + 4)$ cm and $DE = (3y - 6)$ cm. Find x, y, $\angle A$ and AB. If $\angle B = 40^\circ$, find $\angle F$. 5

OR

Pipe A fills a tank in 6 hours, pipe B in 12 hours, and a drain empties the full tank in 24 hours. Find the fraction filled in 1 hour with all three open, and the time to fill the empty tank. 5

END OF QUESTION PAPER — Total: 50 Marks

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